

AfyaSasa — System Audit & Readiness Report

Report date: 15 July 2026 (updated)

Environment: `http://localhost:8080 (app)` · `http://localhost:3000/api/v1 (API)`

Tenant: demo (single-hospital / Jalaram profile)

Primary admin: `it@jalaram.co.ke`

Companion guide: `AfyaSasa-System-Functionality-Guide.pdf` (step-by-step functionality + strength assessment)

1. Executive summary

AfyaSasa is a Docker-hosted hospital EMR covering reception, outpatient, laboratory, imaging, inpatient, emergency, specialty services, reporting, notifications, and hospital administration.

Overall readiness (15 Jul 2026): Suitable for **demo, structured UAT, and controlled clinical pilot**. Production VPS go-live is recommended after TLS, backups, credential rotation, Celcom live keys, and the Priority-1 checklist in §6.

Strength snapshot: Module breadth **high**; core OPD/lab/imaging/IPD/ED journeys **operational**; pharmacy and national multi-tenant schema work remain **foundational / enterprise backlog**.

Delivery since earlier July audit

- Navigation regrouped (Referrals/Sick Sheets → Outpatient; Lab & Imaging dashboards; Nursing nav)
 - Realtime publish expanded (lab/radiology create+status, discharge, ED alerts)
 - Facilities: Jalaram Hospital + City Clinic; ED permissions hardened
 - Audit logs: before/after snapshots + CSV export/import
 - Clinical orders mirror + pharmacy scaffold UI
 - IPD admissions HTTP 500 fixed; IPD / Lab / Imaging dashboards improved
 - Offline catalog cache + mutation queue foundations
 - Jalaram identity emails (`it@jalaram.co.ke` and staff `@jalaram.co.ke`)
 - Celcom Africa SMS gateway + bulk SMS admin UI
 - Login password show/hide; larger animated welcome header; collapse arrows on form/nav sections
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2. Platform & infrastructure

Module	What it does	Status	Recommended improvements
PostgreSQL	Clinical + admin data (demo schema)	Working	Nightly encrypted backups; restore drills
Redis	Cache / BullMQ queues	Working	Cache catalogs + report aggregates
MinIO (S3)	Clinical files & templates	Working	Full file_registry consistency checks
Nginx	UI :8080 + API proxy	Working	TLS termination on VPS
NestJS API	REST /api/v1 , Swagger /docs	Working	Uniform pagination + error codes
React frontend	Clinical SPA + Control Center	Working	Code-split large bundle; device matrix QA
Realtime (Socket.IO)	Live worklist / ED / settings sync	Working	Authenticate socket handshake with JWT
Docker Compose	Pilot packaging	Working	Production compose + resource limits

3. Core platform services

Module	What it does	Status	Recommended improvements
Authentication	Login, refresh, logout, forced password change, show/hide password	Working	Email reset in production; MFA for admins
RBAC	Roles, permissions, revocation	Working	Job-title role templates
Tenancy / facilities	X-Tenant + facilities JSON (Jalaram + City Clinic)	Working for single hospital	Multi-schema only when expanding customers
Admin / Settings	Profile, catalogs, departments	Working	Catalog change history
Audit logs	Mutation trail + export/import	Working	SIEM shipping optional later
Notifications / SMS	Inbox + Celcom/AT/Twilio + bulk SMS UI	Working (stub until keys)	Live Celcom credentials + delivery dashboard
Clinical orders mirror	Lab/imaging/pharmacy unified view	Partial	Deeper pharmacy dispense
Worklists	Cross-department queues	Working	Saved filters per user
Documents / templates	DOCX / library / medical docs	Working	PDF parity for every template

Health check	/api/v1/health	Working	Deep dependency latency probes
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4. Clinical modules (main navigation)

4.1 Reception

Screen	Explanation	Status	Improvements
Patient Search	Find patient; open safety banner/profile	Working	National ID / duplicate scoring
Register Patient	Progressive registration; collapsible optional steps	Working	Photo capture; CSV import
Patient Timeline	Chronological journey	Working	Timeline PDF export
OPD Check-In	Patient → clinic → visit type → confirm	Working	Appointment auto check-in
Appointments	Schedule + status workflow	Working	SMS reminders

4.2 Outpatient

Screen	Explanation	Status	Improvements
Triage Queue	Vitals, colour triage, alerts	Working	NEWS2/MEWS auto-score
OPD Patients	Clinic patient directory	Working	Room/panel view
Doctor Queue	SOAP, diagnosis, complete visit	Working	Order sets; e-Rx
Referrals	Referral letters + tracking	Working	External eReferral
Sick Sheets	Issue/print/store certificates	Working	Employer QR verify

4.3 Documents

Screen	Explanation	Status	Improvements
Medical Documents	Patient document repository	Working	OCR search
Hospital Library	Policies / SOPs / circulars	Working	Staff acknowledgement tracking

4.4 Laboratory

Screen	Explanation	Status	Improvements
Lab Dashboard	Pending/TAT command view	Working	TV wallboard mode
Laboratory	Request → sample → result → verify	Working	HL7 analyzer feed

Lab Patients	Active lab patients	Working	Batch collection
Results Inbox	Clinician verified inbox	Working	Acknowledge receipts
Lab catalog (admin)	Panels/tests + CSV import	Working	LOINC + pricing

4.5 Imaging

Screen	Explanation	Status	Improvements
Imaging Dashboard	Pending studies overview	Working	Modality utilisation charts
Radiology	Worklist, report, PDF attach	Working	DICOM / PACS
Imaging Patients	Active imaging patients	Working	Room assignment
Radiology catalog	Modalities + studies CSV import	Working	Keep study templates current

4.6 Inpatient

Screen	Explanation	Status	Improvements
Inpatient (IPD)	Ward board, beds, workspace	Working	Visual bed map designer
IPD Patients	Admissions directory	Working	Expected discharge dates
ICU / HDU	Critical / step-down views	Working	Ventilator / hourly grids
Nursing	Nursing command (notes/obs)	Working	Full MAR UI

4.7 Emergency

Screen	Explanation	Status	Improvements
Emergency	Queue, bays, metrics, alerts + sound	Working	Dedicated ED TV mode
ED Patients	ED directory	Working	Ambulance pre-alert

4.8 Specialty

Screen	Explanation	Status	Improvements
Theatre	Bookings / procedures	Working	WHO checklist
Maternity	ANC → labour → delivery → nursery	Working	Partograph
Pharmacy	Orders / dispense pilot	Partial	Stock & batches

4.9 Reports & administration

Screen	Explanation	Status	Improvements
Executive Analytics	BI KPIs, trends, widgets, CSV	Working	Interactive drill-down

Operations Center	Today's ops command view	Working	Configurable refresh
Clinical Reports / OPD Reports	Clinical & OPD reporting	Working	Scheduled emails
Clinical Orders	Unified order board	Working	Closer pharmacy loop
Worklists	Cross-department queues	Working	Role landing pages
Notifications	Staff inbox	Working	Priority filters
Hospital Control Center	Full hospital OS	Working	Guided go-live wizard

5. Control Center modules

Section	Status	Notes
Hospital profile	Working	Drives letterhead / branding defaults
Facilities & modules	Working	Jalaram + City Clinic pattern
User & access (Account Center)	Working	Doctor quick-add, specialisation, reset password
Roles & permissions	Working	Fine-grained RBAC
Departments & clinics	Working	Feeds OPD routing
Clinical / Lab / Radiology catalogs	Working	CSV import for lab & radiology
Wards & beds	Working	IPD occupancy source
Theatre / Maternity config	Working	Specialty catalogs
Templates / Printing / Branding	Working	Identity package
Notification center (bulk SMS)	Working	Celcom-ready
Audit / Security / Backup / Health	Working	Ops governance
Super admin	Working	Restrict tightly in production

6. Improvement roadmap

Priority 1 — Before broader pilot users

1. Create real clinical users in Account Center (roles + specialisation)
2. Import radiology study protocols CSV if empty
3. Complete one full UAT journey with director sign-off
4. Rotate demo passwords for any shared accounts
5. Keep production secrets out of chat logs / git

Priority 2 — Before Truehost / Contabo VPS go-live

- 1. TLS + HTTPS-only reverse proxy
- 2. Nightly Postgres backup + restore test
- 3. Celcom Africa live keys smoke-test
- 4. Firewall / SSH hardening / non-root deploy user
- 5. Resource sizing: 4 vCPU / 8 GB RAM / 80+ GB SSD

Priority 3 — Product depth

- 1. Pharmacy warehouse and MAR
- 2. MFA for admins; email reset
- 3. Socket JWT auth
- 4. Chart drill-down for Executive Analytics
- 5. Maternity partograph / theatre WHO checklist

Priority 4 — Enterprise scale (later)

- 1. Multi-schema ORM (remove hardcoded demo schema)
- 2. Redis report/catalog cache
- 3. Deep entity audit snapshots
- 4. HL7 / PACS integrations
- 5. National report auto-submission

Hosting recommendation: Prefer **Truehost (Kenya)** for Jalaram primary production latency and local support; Contabo as optional secondary / DR capacity.

7. Automated / previous test posture

Earlier automated run (9 Jul 2026): **43 PASS · 3 WARN · 0 FAIL.**

Notable WARN items since addressed or clarified:

Previous WARN	Current posture
IPD admissions HTTP 500	Fixed (listAdmissions where clause)
ED bays 403 for admin	Expected without ED bay perms; admin ED perms improved via migration
Empty radiology study protocols	Resolved by CSV import when run; keep catalog maintained

Re-run `ops/full-system-audit.sh` after major releases for an updated pass matrix.

8. Manual audit checklist (director sign-off)

- ☐ Login as `it@jalaram.co.ke` — hospital code hidden; password toggle works
- ☐ Welcome header visible and animated
- ☐ Reception — register → check-in → timeline
- ☐ OPD — triage → doctor queue → complete
- ☐ Lab — order → result → Results Inbox
- ☐ Radiology — request → report → verify
- ☐ IPD — admit → ward board loads; nursing notes
- ☐ ED — red triage → alert visible
- ☐ Account Center — add doctor with specialisation
- ☐ Radiology catalog CSV import verified
- ☐ Executive Analytics date range + CSV export
- ☐ Notifications bulk SMS page opens (stub OK)
- ☐ Collapse arrows shrink form / nav sections
- ☐ Mobile drawer nav groups collapse

9. How to open these reports

File	Purpose
<code>AfyaSasa-System-Functionality-Guide.pdf</code>	Detailed step-by-step functionality + strength assessment
<code>AfyaSasa-System-Audit-Latest.pdf</code>	This status/audit report
Matching <code>.html</code> / <code>.md</code>	Browser-friendly / editable sources

Generate PDFs:

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``bash

bash ops/generate-audit-pdf.sh ops/audit-reports/AfyaSasa-System-Functionality-Guide.md
bash ops/generate-audit-pdf.sh ops/audit-reports/AfyaSasa-System-Audit-Latest.md
...

```

AfyaSasa system audit — Jalaram Hospital profile — updated 15 July 2026.

AfyaSasa · Jalaram Hospital profile · Generated from markdown · Confidential hospital operations document